



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY
A.S Nagar, Elupatti– Thanjavur.
 (Approved by AICTE-New Delhi, Affiliated to Anna University, Chennai.)

Department of Mechanical Engineering

CONTINUOUS INTERNAL ASSESSMENT TEST-II, OCT - 2026

Reg.No	8	2	1	9							
Regulation	R2021										
Course Code-Name	OML351 – Introduction to Non-Destructive Testing								SET- B		
Year / Semester	IV & VII								Total Marks:100		
Date - Time	__.10.2026 / 9.30AM TO 12.30PM										
Course Instructor	Dr. S. R. Sathishkumar, AP/ MECH										
C406-OEIII14.3 - Calibrate the instrument and inspect for in-service damage in the components by means of Eddy current testing as well as Thermography testing.											
C406-OEIII14.4 - Differentiate various techniques of UT and AET and select appropriate NDT methods for better evaluation.											
C406-OEIII14.5 - Interpret the results of Radiography testing and also have the ability to analyse the influence of various parameters on the testing.											
L1-Remember , L2-Understand , L3-Apply , L4-Analyze , L5 -Evaluate , L6-Create											

Answer All Questions / PART – A (10×2 = 20 Marks)

Q.NO	QUESTIONS	Marks	CO	LEVEL
1.	List the primary instruments used for infrared detection in thermography.	2	C406-OEIII14.3	R
2.	Differentiate between contact and non-contact methods of thermography.	2	C406-OEIII14.3	U
3.	Explain the limitations of the eddy current testing method.	2	C406-OEIII14.3	U
4.	Name the main components of a basic acoustic emission instrumentation chain.	2	C406-OEIII14.4	R
5.	List two industrial applications of the acoustic emission technique.	2	C406-OEIII14.4	R
6.	Compare A-scan, B-scan and C-scan data presentation methods.	2	C406-OEIII14.4	U
7.	Compare the advantages and limitations of ultrasonic testing and acoustic emission technique.	2	C406-OEIII14.4	U
8.	State the purpose of an exposure chart in radiography.	2	C406-OEIII14.5	R
9.	Differentiate between the generation of X-rays and gamma rays.	2	C406-OEIII14.5	U
10.	Explain the importance of the inverse square law in radiation safety for personnel.	2	C406-OEIII14.5	U

PART – B (5×13 = 65 Marks)

11.a	Apply infrared thermography instrumentation to suitable applications, with a neat block diagram.	13	C406-OEIII14.3	Ap
OR				
11.b	Apply the following terms to an eddy current inspection with practical examples: (i) Inductive reactance (ii) Impedance (iii) Lift-off (iv) Edge effect.	13	C406-OEIII14.3	Ap
12.a	Apply AE instrumentation to design an experimental setup for acoustic emission testing and explain the function of each component.	13	C406-OEIII14.4	Ap
OR				
12.b	Apply TOFD and phased array techniques in ultrasonic testing to suitable inspection requirements, with neat sketches.	13	C406-OEIII14.4	Ap
13.a	(i) Apply A, B and C-scan presentations to specific engineering applications. (ii) Demonstrate the construction of an ultrasonic probe with a neat diagram.	13	C406-OEIII14.4	Ap
OR				
13.b	Apply different methods of ultrasonic testing to real-world inspection problems, with suitable sketches.	13	C406-OEIII14.4	Ap
14.a	Demonstrate the generation of gamma rays by radioactive decay and apply their key properties to industrial radiography.	13	C406-OEIII14.5	Ap
OR				
14.b	Apply the inverse square law and exposure factors (kV, mA, time, SFD, thickness) to select radiographic technique parameters.	13	C406-OEIII14.5	Ap
15.a	Apply SWSI and DWSI techniques in radiography testing of pipes, with neat sketches.	13	C406-OEIII14.5	Ap
OR				
15.b	Apply a standard film radiography workflow to minimise handling and processing defects.	13	C406-OEIII14.5	Ap
PART – C (1×15 = 15 Marks)				
16.a	Evaluate eddy current testing and its instrumentation for a specific engineering application, and assess its limitations and improvements.	15	C406-OEIII14.3	E
OR				
16.b	Evaluate active and passive thermography for a given inspection scenario and justify the most effective method.	15	C406-OEIII14.3	E

Course Instructor

HOD

IQAC Coordinator